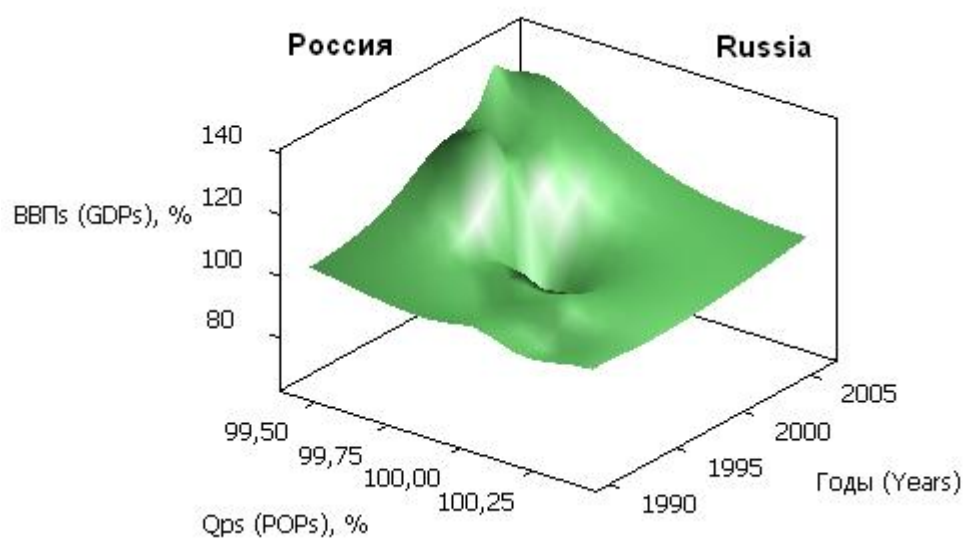


Э.А. ПИЛЬ

**ТЕОРЕТИЧЕСКИЕ И СТАТИСТИЧЕСКИЕ ВАРИАНТЫ
РАЗВИТИЯ ЭКОНОМИКИ И НАСЕЛЕНИЯ
РАЗЛИЧНЫХ СТРАН МИРА И ИХ ПРОГНОЗ**

**THEORETICAL AND STATISTICAL VARIANTS OF
DEVELOPMENT OF ECONOMY AND POPULATION IN
DIFFERENT COUNTRIES AND THEIR PREDICTION**

Книга 1



Санкт-Петербург
2011

'For me a word of wisdom is more valuable than any gold.'

Democritus

INTRODUCTION

The calculation and prediction of GDP, the primary indicator of an economy, as well as the estimation of population size, both for individual countries and the world as a whole, are the subject of numerous monographs and articles. Consequently, many people constantly face the question of what direction their own country's economy, and the global economy in particular, will take in the near future, and how this will affect the income of the average citizen.

Based on the analysis of statistical data variables, the author has derived polynomial dependencies with sufficiently high correlation coefficients for most of them, which allows for the prediction of the considered variable values with a high degree of accuracy.

This book is dedicated to my son, *Mikhail*, and to the memory of my parents and elder brother.

CONTENTS

VOLUME I

INTRODUCTION	3
CHAPTER 1	
SOME QUESTIONS ABOUT ECONOMY, POPULATION AND THEIR PREDICTIONS	5
1.1. Parameters Characterizing a Country's Economy	5
1.2. Demography of Various Countries of the World	23
1.2.1. World Population Growth	23
1.2.2. Demographic Indicators	27
1.2.3. General Theory of Human Growth	33
1.2.4. Demographic Transition	34
1.2.5. World Demographic Transition	36
1.3. Essence and Classification of Predictions	37
1.3.1. Essence of Financial Predictions	37
1.3.2. Classification of Financial Predictions	42
1.4. Prediction Systems	45
1.4.1. Types of Prediction Systems, Typical Structure	45
1.4.2. Organization of Prediction	48
1.4.3. The Prediction and Planning System of the Russian Federation	52
1.5. Macro- and Micro Predictions	54
1.5.1. Prediction of Macroeconomic Indicators	54
1.5.2. Prediction of Socio-Economic Indicators	59
1.5.3. Demand Prediction	61
CHAPTER 2	
THEORETICAL KINDS OF CHANGING IN ECONOMY AND POPULATION	66
2.1. Construction of a Table of Scenarios for Economic and Population Development	66
2.2. Scenarios for the Development of Variables <i>et</i> and <i>pt</i>	75
2.3. Classification of the Development of Variables <i>et</i> and <i>pt</i>	118
2.4. Classification of Countries by Area Occupied by GDP and Population	119
CONCLUSION	123
CHAPTER 3	
THEORETICAL OSCILLATIONS OF ECONOMY AND POPULATION	124
3.1. Damped Oscillations of Theoretical Population Stabilization	124
3.2. Practical Change in a Country's Population	128
Small Oscillations Around a Given State	130
Small Oscillations and Stability	137
3.3. Theoretical Scenarios of Decay Time <i>Tod</i> Development Depending on Mortality <i>Dr</i> , Birth Rate <i>Fr</i> and Population <i>Po</i>	140
3.4. Practical Scenarios of Decay Time <i>Tod</i> Development Depending on Mortality <i>Dr</i> , Birth Rate <i>Fr</i> and Population <i>Po</i>	181
3.4.1. General Patterns of Decay Time <i>Tod</i> Development	181
3.4.2. Decay Time <i>Tod</i> by Countries	191
Countries G8	191
USA	191
Japan	195
Germany	199
France	203
Italy	207
UK	211
Canada	215
Russia	219
Countries BRIC	223
People's Republic of China	223
Brazil	227
India	231
Developed countries	235
Sweden	235
Finland	239
Spain	243
Korea, South	247

Australia	251
Turkey	255
Developing countries	259
Nigeria	259
Zambia	263
Peru	267
Countries that Experienced Natural Disasters or Government Change	271
Fiji	271
Indonesia	275
the Dominican Republic	279
Haiti	283
CONCLUSION	287
CHAPTER 4	
ANALYSIS OF GDP AND POPULATION OF DIFFERENT COUNTRIES BASED ON DERIVED PARAMETERS	288
4.1. Countries G8	288
USA	290
Japan	299
Germany	306
France	312
Italy	318
UK	324
Canada	330
Russia	337
4.2. Countries BRIC	343
People's Republic of China	343
Brazil	349
India	354
4.3. Developed countries	361
Sweden	361
Finland	366
Spain	372
Korea, South	378
Australia	384
Turkey	390
4.4. Developing countries	396
the United Arab Emirates	396
Nigeria	402
Zambia	408
Peru	414
4.5. Countries that Experienced Natural Disasters or Government Change	420
Fiji	420
Indonesia	426
the Dominican Republic	433
Haiti	439
4.6. Summary Tables and Figures	446
Summary Tables by Descending Correlation Coefficient	446
Summary Tables by Descending Mean Values	451
CONCLUSION	457